

DATA-260 HW3

Rachandeep Kaur

Github Link- <https://github.com/rachandeepkaur/data260-7117.git>

Step 0: Personal Configuration

Value	Definition	Result
SID4	Last four digits of SJSU Student ID	7117
PORT_BASE	$8000 + (\text{SID4} \bmod 900)$	8817
PREFIX	"s" + SID4	s7117
SEED	SID4	7117
VERIFY_SEED	$260000 + \text{SID4}$	267117
DOMAIN_ID	$\text{SID4} \bmod 8$	5 (Local restaurant inspection)

Hardware: MacBook Pro, 16GB RAM

Model: qwen3:8b, all-MiniLM-L6-v2 (ONNX, via fastembed)

Commit hash:

1. Routes - Handled Separately in a router:

- Home Page (/)

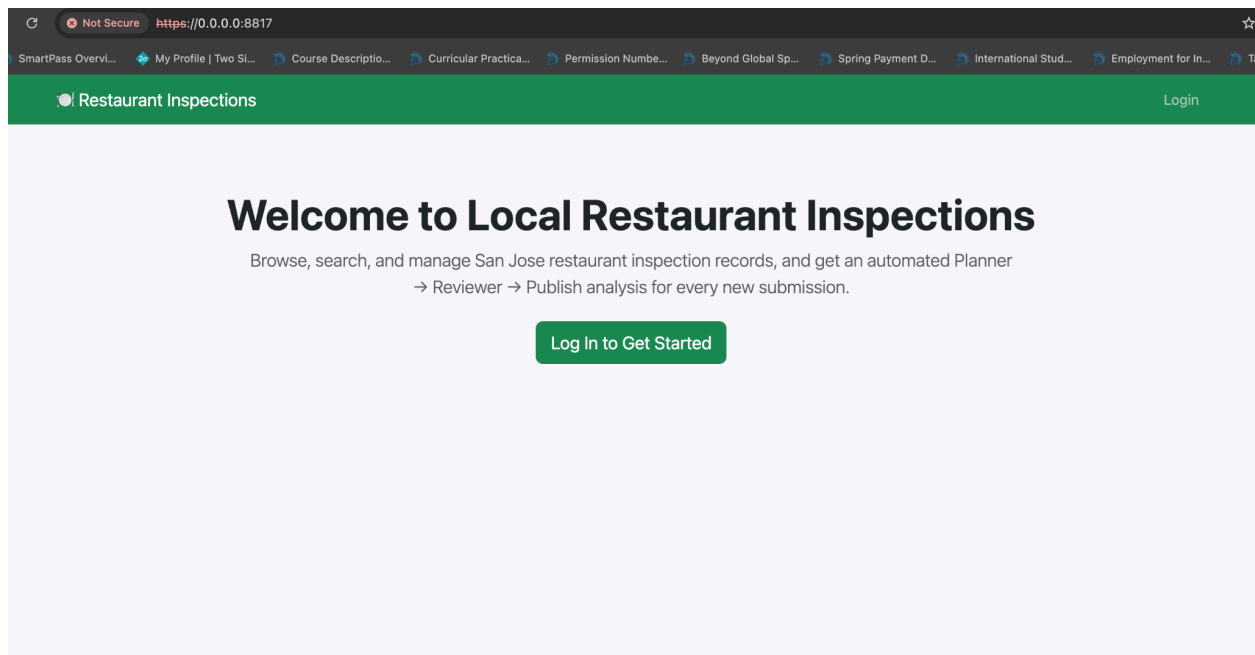
code > web_application > templates > <> index.html > ...

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Home - Rachandeep Kaur</title>
7    <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10
11    {% set active_page = "home" %}
12    {% include "_navbar.html" %}
13
14    <main class="home-hero">
15      <h1>Welcome to Local Restaurant Inspections</h1>
16      <p class="home-subtitle">
17        Browse, search, and manage San Jose restaurant inspection records, and
18        get an automated Planner &rarr; Reviewer &rarr; Publish analysis for
19        every new submission.
20      </p>
21      <div class="home-cta">
22        {% if user %}
23          <a class="home-cta-btn" href="/dashboard">Go to Dashboard</a>
24        {% else %}
25          <a class="home-cta-btn" href="/login">Log In to Get Started</a>
26        {% endif %}
27      </div>
28    </main>
29
30  </body>
31  </html>
32
```

```

@router.get("/")
@router.get("/home")
def home(request: Request):
    """
    Home page route.
    - Registered at both "/" (the assignment's required path) and "/home"
      (an alias) - both render the same page.
    - Checks if a user is logged in using the session
    - Passes user info to the template, which shows a Login link when
      logged out, or a Dashboard + Logout link when logged in
    """
    user = request.session.get("user")
    return templates.TemplateResponse(
        request,
        "index.html",
        {
            "user": user
        }
    )

```



• Login Page (/login)

code > web_application > templates > <> login.html > ...

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Login - Rachandeep Kaur</title>
7   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
8   <link rel="stylesheet" href="style.css">
9 </head>
10 <body>
11
12   {% set active_page = "login" %}
13   {% include "_navbar.html" %}
14
15   <main class="auth-card">
16     <h1>Log In</h1>
17
18     {% if error %}
19     <div class="alert alert-danger" role="alert">Invalid username or password.</div>
20     {% endif %}
21
22     <!-- Classic Post/Redirect/Get: the browser posts this form directly to
23      /login (see routers/auth.py), which redirects to /dashboard on
24      success or back here with ?error=1 on bad credentials. -->
25     <form method="post" action="/login">
26       <div>
27         <label for="username">Username</label>
28         <input type="text" id="username" name="username" placeholder="admin" autofocus required>
29       </div>
30       <div>
31         <label for="password">Password</label>
32         <input type="password" id="password" name="password" placeholder="*****" required>
33       </div>
34       <button type="submit">Log In</button>
35     </form>
36     <p class="auth-note">Demo credentials: <strong>admin</strong> / <strong>password</strong></p>
37   </main>
38
39 </body>
```

```

6
7 @router.get("/login")
8 def login_page(request: Request):
9     """
10    Displays the login form.
11    If the user is already logged in,
12    the template can choose what to display.
13    """
14    user = request.session.get("user")
15    return templates.TemplateResponse(
16        request,
17        "login.html",
18        {
19            "user": user,
20            # Set on the redirect back here after a failed attempt so the template can
21            "error": request.query_params.get("error"),
22        }
23    )
24

```

→ [Not Secure https://0.0.0.0:8817/login](https://0.0.0.0:8817/login)

SmartPass Overvi... My Profile | Two Si... Course Descriptio... Curricular Practica... Permission Numbe... Beyond Global Sp... Spring Payment D... International Stud... Employment for In...

Restaurant Inspections [Login](#)

Log In

Username

Password

[Log In](#)

Demo credentials: **admin / password**

Error screenshot

Not Secure https://0.0.0.0:8817/login?error=1

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Restaurant Inspections Login

Log In

Invalid username or password.

Username

Password

Log In

Demo credentials: **admin** / **password**

Dashboard Page (/dashboard)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <!-- Page title with HW2 - {Name} -->
  <title>HW2 - Rachandeep Kaur</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  {% set active_page = "dashboard" %}
  {% include "_navbar.html" %}

  <p class="welcome-message">Welcome, {{ user }}!</p>

  <!-- Heading naming assigned domain entity using <h1> (biggest font size) -->
  <h1>Local Restaurant Inspections</h1>

  <!-- Form for domain entity -->
  <form id="inspectionForm">
```

```
@router.get("/dashboard")
def dashboard(request: Request):
    """
    Protected route.
    - Only accessible if user is logged in
    - Redirects to login page if session is missing
    """
    user = request.session.get("user")
    # If user is not logged in, block access
    if not user:
        return RedirectResponse(
            url="/login",
            status_code=HTTP_302_FOUND
        )

    # If user is logged in, render dashboard
    return templates.TemplateResponse(
        request,
        "dashboard.html",
        {
            "user": user
        }
    )
```

Not Secure https://0.0.0.0:8817/dashboard

SmartPass Overvi... My Profile | Two Si... Course Descriptio... Curricular Practica... Permission Numbe... Beyond Global Sp... Spring Payment D... International Stud...

Restaurant Inspections Dashboard Log Out

Welcome, admin!

Local Restaurant Inspections

Facility Name / ID:

FA0206933 - YUMMY KITCHEN

Program Site Address:

1711 BRANHAM LN A9, SAN JOSE, CA 95118

Submitter Email:

inspector@countyhealth.gov

Inspection Findings Summary:

Provide inspection observations and compliance details (must be > 25 characters)...

Program Element / Category:

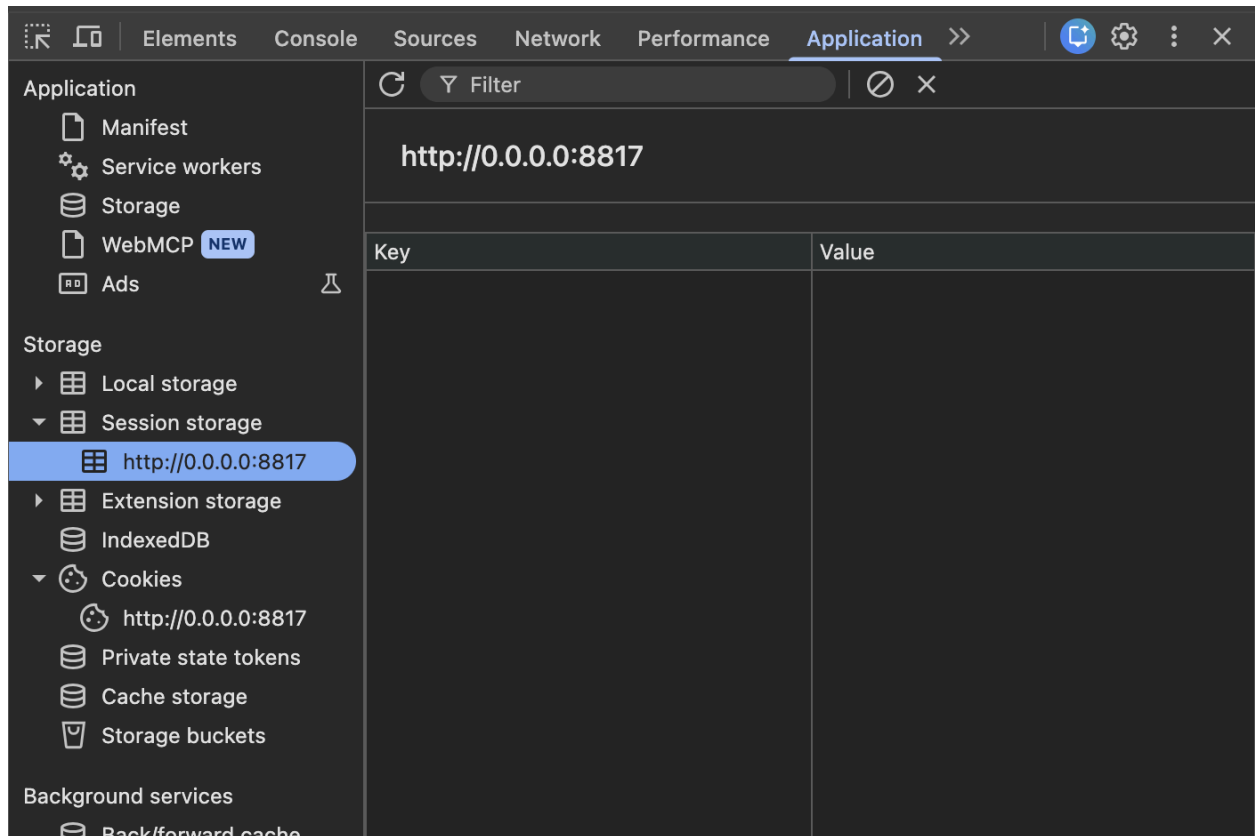
-- Select Inspection Category --

☐ I agree to the terms and conditions.

Submit Inspection Report

Logout (/logout):

```
@router.get("/logout")
def logout(request: Request):
    """
    Logs the user out.
    - Clears all session data
    - Redirects back to home page
    """
    request.session.clear()
    return RedirectResponse(
        url="/",
        status_code=HTTP_302_FOUND
    )
```



Welcome to Local Restaurant Inspections

Browse, search, and manage San Jose restaurant inspection records, and get an automated Planner → Reviewer → Publish analysis for every new submission.

[Log In to Get Started](#)

2. Session Management:

1. Use Starlette's SessionMiddleware to manage user sessions. Store the logged-in user's information in the session.

```
from starlette.middleware.sessions import SessionMiddleware

THIS_DIR = Path(__file__).resolve().parent
CODE_DIR = THIS_DIR.parent

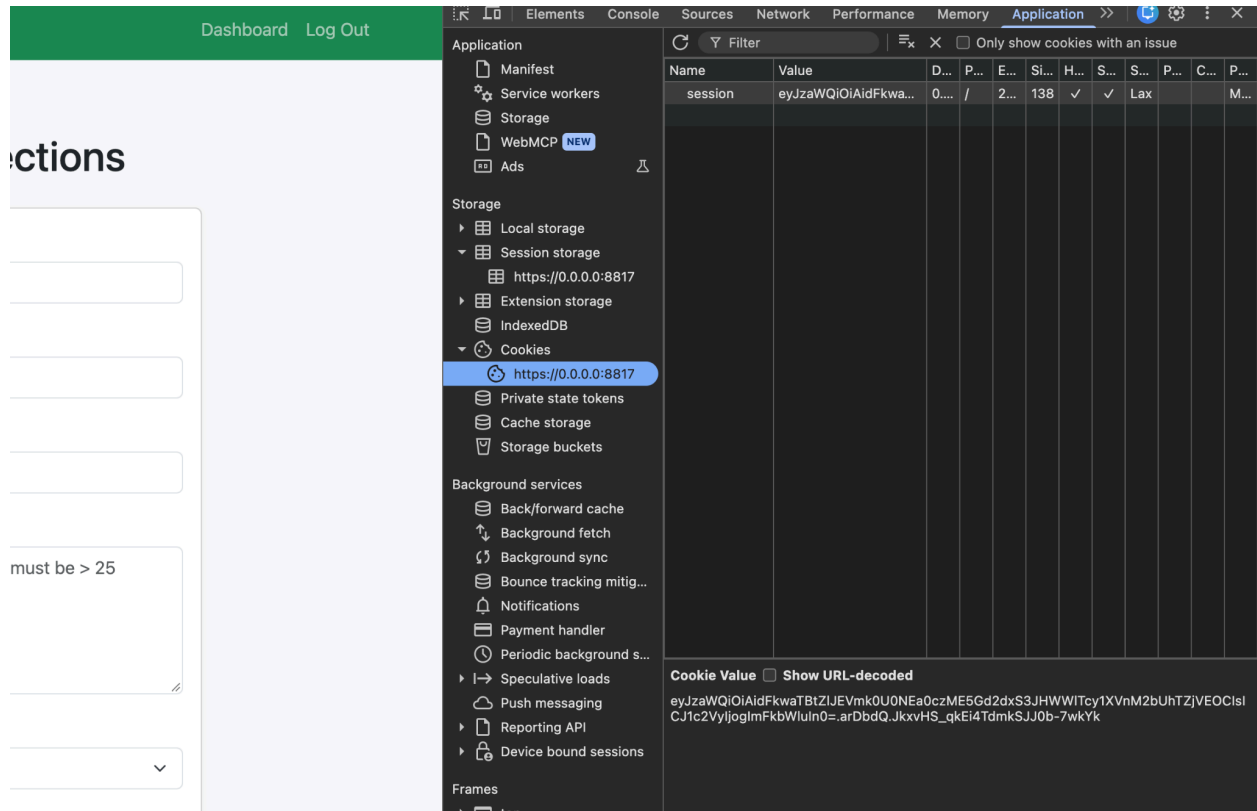
sys.path.insert(0, str(THIS_DIR))
sys.path.insert(0, str(CODE_DIR))
from agents_demo import (
    InspectionSubmission,
    ModelClient,
    ModelClientError,
    PublishOutput,
    run_pipeline,
)
from routers.auth import router as auth_router
from routers.home import router as home_router

# Create FastAPI app
app = FastAPI(title="Restaurant Inspection Records API")

# Local dev only - the frontend and API run on different origins/ports here
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_methods=["*"],
    allow_headers=["*"],
)

# Secret key for session signing - set a real one via env in any deployment
# once other people know it.
SECRET_KEY = os.getenv("SECRET_KEY", "dev-only-secret-key")

# Enable session support
app.add_middleware(
    SessionMiddleware,
    secret_key=SECRET_KEY,
    https_only=True,
    same_site="lax",
    max_age=3600,
)
```



2. Ensure that the session cookie is secure. Capture and include the Set-Cookie response header showing all three attributes.

```
# Enable session support
app.add_middleware(
    SessionMiddleware,
    secret_key=SECRET_KEY,
    https_only=True,
    same_site="lax",
    max_age=3600,
)
```

```

(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % curl -ik -c jar.txt -d "username=admin&password=password" https://localhost:8817/login
HTTP/1.1 302 Found
date: Mon, 21 Sep 2026 07:36:31 GMT
server: uvicorn
content-length: 0
location: /dashboard
vary: Cookie
set-cookie: session=eyJzaWQ1OiAiaFU2T1RIOHVZU2pLbG0wRTBKdEV1cFNsSGNSVnJYNGlMR3JPakltZ084OCIsICJ1c2VyIjogImFkbWluIn0=.arDefw.G552qphzwDR0r1_kksRyEB2xJgQ; path=/; Max-Age=3600; samesite=lax; secure

(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % curl -ik -b jar.txt https://localhost:8817/dashboard
HTTP/1.1 200 OK
date: Mon, 21 Sep 2026 07:36:50 GMT
server: uvicorn
content-length: 8191
content-type: text/html; charset=utf-8
vary: Cookie

<!DOCTYPE html>
<html lang="en">

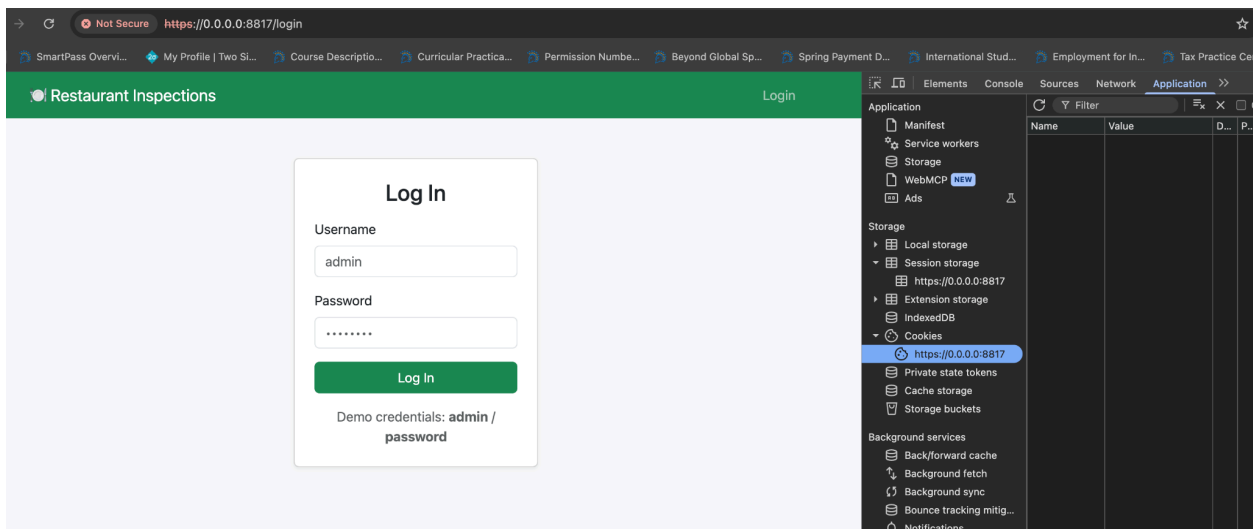
```

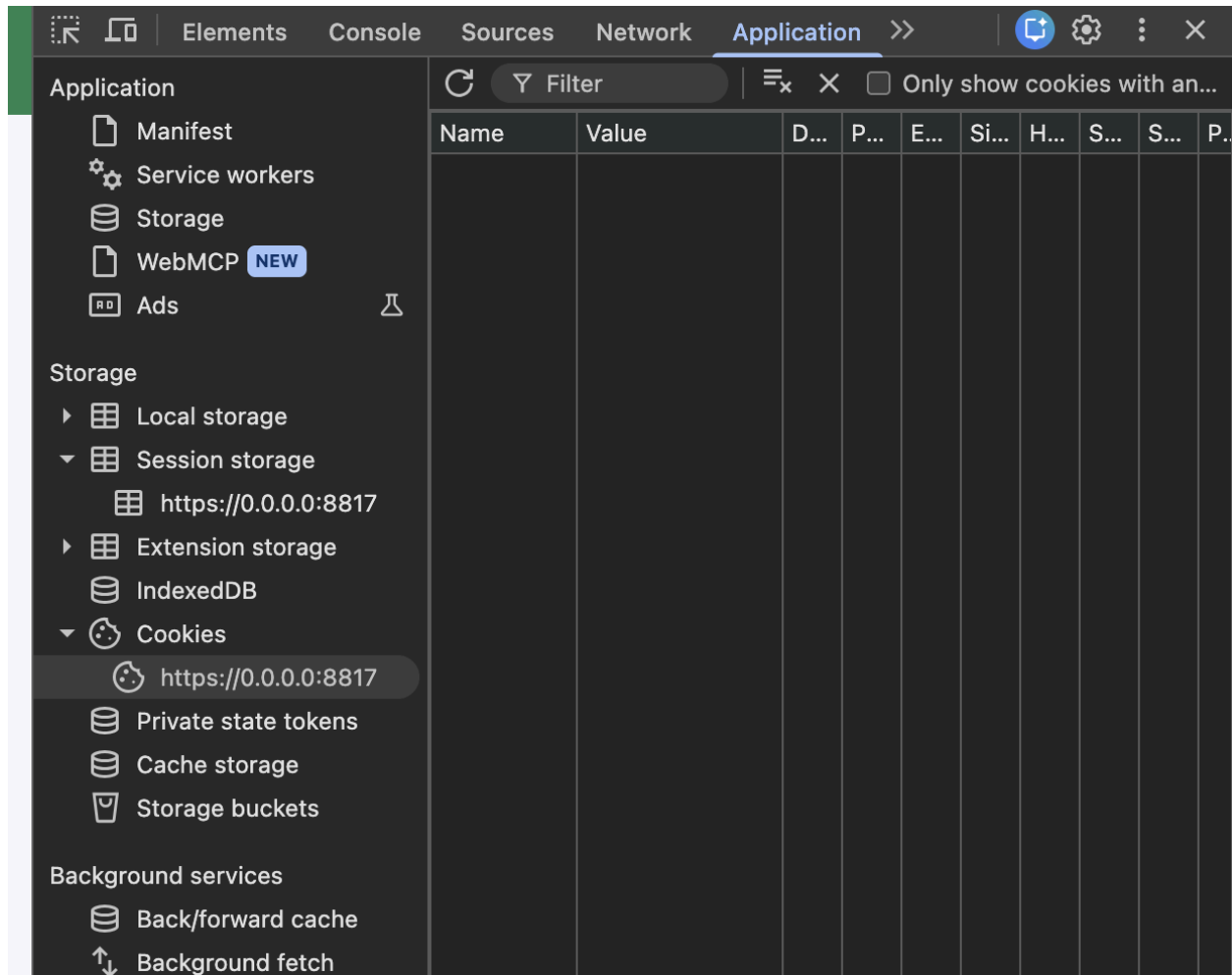
3. Implement an idle session timeout. Prove that an expired or logged-out session cannot be reused to reach /dashboard

```

(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % cd code/web_application
(.venv) (base) pc@pcs-MacBook-Pro web_application % curl -ik -b old.txt https://localhost:8817/dashboard
HTTP/1.1 302 Found
date: Mon, 21 Sep 2026 09:18:16 GMT
server: uvicorn
content-length: 0
location: /login
vary: Cookie

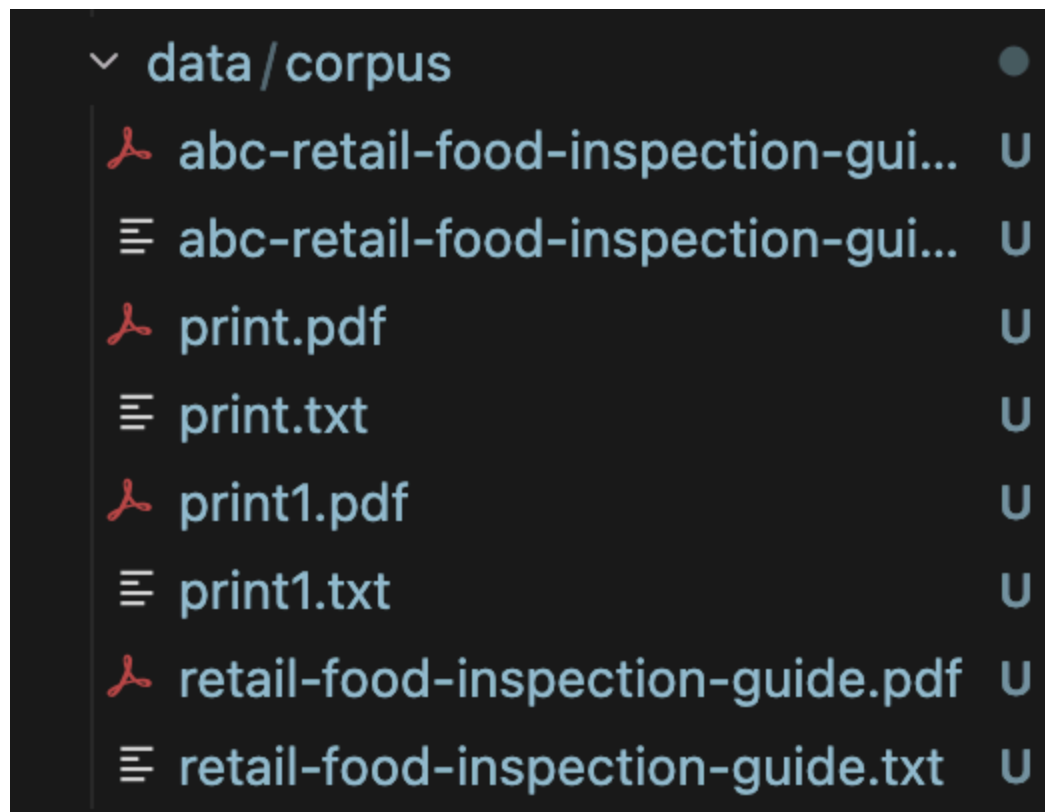
```





Part 2: Compare Three LlamaIndex Chunking Techniques (Retrieval-Only RAG)

Dataset



Size of the dataset

```
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % du -ch data/corpus/*.txt | tail -1  
592K    total  
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 %
```

reports > hw03 > {} CORPUS_MANIFEST.json > ...

```
1  [
2    {
3      "file": "abc-retail-food-inspection-guide.pdf.pdf",
4      "bytes": 758993,
5      "sha256": "eea6af4a62b1de5227d8fd8a859aeceb0e89a4cad98d9b911c12dc93642c86d"
6    },
7    {
8      "file": "abc-retail-food-inspection-guide.pdf.txt",
9      "bytes": 146266,
10     "sha256": "82499fc903f352197233efed6cee629e930577abc6a37b3041efb9563e07da71"
11   },
12   {
13     "file": "print.pdf",
14     "bytes": 310299,
15     "sha256": "45819ad1e2e0dc052f7df1056e1797d312567bdb6ecffa84c3230dbea915dd4f"
16   },
17   {
18     "file": "print.txt",
19     "bytes": 3316,
20     "sha256": "e13c7244dc438a48136aff7b18e7a5f900bcef4942956f8361d9a2c2a593e8b9"
21   },
22   {
23     "file": "print1.pdf",
24     "bytes": 548119,
25     "sha256": "01340fbc81b96edfd00906916258533c81c8e63a0b3799a206da19782afa43bc"
26   },
27   {
28     "file": "print1.txt",
29     "bytes": 6184,
30     "sha256": "477c8d958bc2c52e688c966042093665489a57466d84543d091d263a07b3b342"
31   },
32   {
33     "file": "retail-food-inspection-guide.pdf",
34     "bytes": 2263552,
35     "sha256": "bd8ad5210495705c50387a9d22da09d211f14f2c48de9f8328d79870bb123396"
36   },
37   {
38     "file": "retail-food-inspection-guide.txt",
39     "bytes": 444358,
40     "sha256": "794a183b446ec41cdece9a66e79ee1ce6d48a6480fd8f7d271fc6b9bfcd09ccd"
41   }
42 ]
43
```

HW03 — Domain Corpus Sources

Access dates are based on each file's local download timestamp (all four were fetched 2026-09-21).

#	Source URL	Access date	Local filename	Notes
1	https://admin.publichealth.lacounty.gov/eh/docs/inspection/retail-food-inspection-guide.pdf	2026-09-21	retail-food-inspection-guide.pdf	"Retail Food Inspection Guide" (Oct 2025), LA County Dept. of Public Health, Environmental Health
2	https://sbcounty.gov/uploads/dph/ehs/Programs/Food/FoodFacilities/abc-retail-food-inspection-guide.pdf	2026-09-21	abc-retail-food-inspection-guide.pdf.pdf	"ABC Retail Food Inspection Guide" (2018), San Bernardino County Public Health, Environmental Health Services
3	https://inspections.myhealthdepartment.com/sacramento	2026-09-21	print.pdf	Single facility inspection report printout - Pioneer House (FA0005532), Sacramento
4	https://inspections.myhealthdepartment.com/sacramento	2026-09-21	print1.pdf	Single facility inspection report printout - Dat Thanh Restaurant (FA0003611), Sacramento

Questions yaml file and pushed before retrieval results

```
> hw03 > ! questions.yaml
questions:
- id: q1
  type: single_source
  source_documents:
    - retail-food-inspection-guide.txt
  text: >
    According to the LA County Retail Food Inspection Guide, what score range earns a food facility a "B" letter grade, and what happens
    instead of a letter grade if the score is 69 or below?
  expected_answer: >
    A "B" Letter Grade is issued for a final routine inspection score between 80-89. A score of 69 or below does not get a letter grade at
    all - it gets a "Score Card" instead.
  answer_keyword: "69 or below"
  grounding_location: "retail-food-inspection-guide.txt, line 162"
  grounding_excerpt: >
    "Placard Type Description "A" Letter Grade Final routine inspection score between 90 - 100 "B" Letter Grade Final routine inspection
    score between 80 - 89 "C" Letter Grade Final routine inspection score between 70 - 79 Score Card Final routine inspection score of
    69 or below"

- id: q2
  type: single_source
  source_documents:
    - abc-retail-food-inspection-guide.pdf.txt
  text: >
    Per the San Bernardino County ABC Retail Food Inspection Guide, what must a food facility that receives a "C" grade do to avoid closure,
    and what phone number should an operator call with questions about an inspection report?
  expected_answer: >
    The violations must be corrected, and the facility must submit a written request for a re-score inspection within 30 days of the
    date on the OIR to avoid closure. Questions about the OIR, inspector actions, or applicable law/policy go to EHS at (800) 442-2283.
  answer_keyword: "442-2283"
  grounding_location: "abc-retail-food-inspection-guide.pdf.txt, lines 18-19 and 130-132"
  grounding_excerpt: >
    "If there are questions regarding the OIR, actions by the health inspector or the department, or any applicable law, regulation, or
    policy, contact EHS at (800) 442-2283." / "C - Violations must be corrected, and - The food facility must submit a written request
    for a re-score inspection within 30 days of the date on the OIR to avoid closure."
```

```

questions:
- id: q3
  type: single_source
  source_documents:
    - print1.txt
  text: >
    In the inspection report for Dat Thanh Restaurant (Est ID FA0003611, 5035 Fruitridge Rd, Sacramento), what specific cooling violation did the inspector observe, and what temperature/time rule does it violate?
  expected_answer: >
    The inspector observed 5-gallon buckets cooling at room temperature measuring 155F, and cooked noodles cooling in the walk-in refrigerator measuring 125F after only 30 minutes. Both violate the rule that potentially hazardous food must be rapidly cooled from 135F to 70F within the first 2 hours, and from 135F to 41F within a total of 6 hours (Cal Code 114002, 114002.1). Both were corrected on site.
  answer_keyword: "155F"
  grounding_location: "print1.txt"
  grounding_excerpt: >
    "9.COOLING METHODS Observations: Observed 5-gallon buckets of both cooling at room temperature, measuring 155F. Provided education on proper cooling methods. Corrected on site. Observed cooked noodles cooling in walk in refrigerator measuring 125F after 30 minutes. Corrected on site. Code Description: All potentially hazardous foods shall be rapidly cooled from 135F to 41F within 6 hours. The decrease in temperature from 135F to 70F shall occur within the first 2 hours."

- id: q4
  type: shared_topic
  source_documents:
    - retail-food-inspection-guide.txt
    - abc-retail-food-inspection-guide.pdf.txt
  text: >
    What are the required hot-holding and cold-holding temperatures for potentially hazardous food (PHF), and what exception lets some foods be held up to 45F?
  expected_answer: >
    PHF must be held at or below 41F (cold holding) or at or above 135F (hot holding). Both guides carry the same exception: certain foods (e.g. raw shell eggs, unshucked live molluscan shellfish, pasteurized milk/milk products in sealed containers, PHF held for dispensing on serving lines/salad bars, PHF in transport or vending machines) may be held between 41F and 45F, and PHF received between 41-45F must be cooled to 41F within 4 hours.
  answer_keyword: "135°F"
  grounding_location: "retail-food-inspection-guide.txt line 462; abc-retail-food-inspection-guide.pdf.txt line 274"
  grounding_excerpt: >
    LA: "IN" shall be marked if PHFs are being held at 41F/45F or below or 135F or above within approved time allowances." / San Bernardino: "Potentially hazardous foods shall be held at or below 41F or at or above 135F." plus "Foods approved for holding between 41F and 45F..."

- id: q5
  type: shared_topic
  source_documents:
    - retail-food-inspection-guide.txt
    - abc-retail-food-inspection-guide.pdf.txt
  text: >
    When are food employees required to wash their hands, and what makes an inability to wash hands especially serious under these guides?
  expected_answer: >
    Per the San Bernardino guide, employees must wash their hands before beginning work, before handling food/equipment/utensils, as often as necessary during food preparation, when switching from raw to ready-to-eat foods, after touching body parts, after using the toilet room, or any time contamination may occur; hands/arms not clean, or not washed when necessary, are MAJOR violations. The LA County guide separately lists "Inability of employees to wash their hands" as one of the imminent health hazards that can get a facility's permit suspended and the facility closed immediately.
  answer_keyword: "Inability of employees to wash their hands"

```

A. Environment and setup

≡ requirements.txt

```
1 langchain-ollama>=0.2
2 pydantic>=2.0
3 fastapi>=0.110
4 uvicorn>=0.29
5 python-multipart>=0.0.9
6 itsdangerous>=2.1
7 jinja2>=3.1
8 langgraph>=0.2
9 pypdf>=5.0
10 llama-index-core>=0.14
11 fastembed>=0.7
12 onnxruntime>=1.20
13 pyyaml>=6.0
14 pandas>=2.0
15
```

```
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/hw03/embeddings.py
384
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 %
```

```

(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % pip list | grep -Ei "llama-index|fastembed|faiss|numpy|pandas|onnxruntime"
fastembed                0.8.0
llama-index-core          0.14.24
llama-index-instrumentation 0.6.0
llama-index-workflows     2.24.1
numpy                     2.5.3
onnxruntime               1.23.2
pandas                    3.0.6
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 %

```

B. One retrieval-only pipeline per technique

```

def build_indexes(docs: list) -> tuple[dict[str, VectorStoreIndex], list[dict]]:
    token_splitter = TokenTextSplitter(chunk_size=MAX_TOKEN_CHUNK_SIZE, chunk_overlap=32)
    semantic_splitter = SemanticSplitterNodeParser(
        buffer_size=1, breakpoint_percentile_threshold=95, embed_model=embed_model
    )
    window_splitter = SentenceWindowNodeParser.from_defaults(window_size=3)

    indexes: dict[str, VectorStoreIndex] = {}
    chunk_stat_rows: list[dict] = []
    for name, splitter in [
        ("token", token_splitter),
        ("semantic", semantic_splitter),
        ("sentence_window", window_splitter),
    ]:
        nodes = splitter.get_nodes_from_documents(docs)
        print(f"[{name}] {len(nodes)} nodes")
        for i, node in enumerate[Any](nodes):
            chunk_stat_rows.append({
                "technique": name,
                "chunk_index": i,
                "chunk_length": len(node.get_content()),
            })
        indexes[name] = VectorStoreIndex(nodes, embed_model=embed_model)
    return indexes, chunk_stat_rows

def cosine_similarity(query_vec: np.ndarray, doc_vecs: np.ndarray) -> np.ndarray:
    if doc_vecs.shape[0] == 0:
        return np.array([])
    query_unit = query_vec / np.linalg.norm(query_vec)
    doc_units = doc_vecs / np.linalg.norm(doc_vecs, axis=1, keepdims=True)
    return doc_units @ query_unit

def retrieve(technique: str, index: VectorStoreIndex, question: dict, k: int) -> list[dict]:
    query = question["text"].strip()
    qid = question["id"]

    query_vec = np.array(embed_model.get_query_embedding(query))
    print(f"[{technique}/{qid}] query embedding dim={query_vec.shape[0]}, first 8={np.round(query_vec[:8], 4)}")

    start = time.perf_counter()
    retriever = index.as_retriever(similarity_top_k=k)
    result_nodes = retriever.retrieve(query)

```

reports > hw03 > raw > chunk_stats.csv > data

	technique	chunk_index	chunk_length
1	token	0	1029
2	token	1	1333
3	token	2	2173
4	token	3	2569
5	token	4	1394
6	token	5	1104
7	token	6	1081
8	token	7	1201
9	token	8	1103
10	token	9	1110
11	token	10	1111
12	token	11	1084
13	token	12	1004
14	token	13	1091
15	token	14	1257
16	token	15	1182
17	token	16	1298
18	token	17	1107
19	token	18	1091
20	token	19	1021
21	token	20	979
22	token	21	1104
23	token	22	1067
24	token	23	1121
25	token	24	1106
26	token	25	1137
27	token	26	1007
28	token	27	1121
29	token	28	1141
30	token	29	960
31	token	30	1032
32	token	31	905
33	token	32	924
34	token	33	876
35	token	34	919
36	token	35	849
37	token	36	974
38	token	37	858
39	token	38	909
40	token	39	828
41	token	40	838
42	token	41	934
43	token	42	806
44	token	43	749
45	token	44	883
46	token	45	830
47	token	46	839

B1. One sample chunk from each technique

```
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/hw03/print_sample_chunks.py q4

=====
TOKEN | question=q4 | chunk_length=1237 chars | cosine=0.7388
=====
the more permissive conditions are for growth.
Hot holding The hot holding of food at 135°F or greater is sufficient to prevent the growth of pathogens and is therefore an effective measure in the prevention of foodborne illness.
Time limits for food preparation For potentially hazardous foods, it is required that the food be returned for refrigeration or hot holding after two hours at room temperature. After two cumulative hours at room temperature some pathogens begin entering the log phase of growth where the multiplication of microbes replicates logarithmically to produce enough toxins or pathogens to cause illness if ingested.
Receiving temperatures Temperature is one of the prime factors that control the growth of bacteria in food. Many, though not all, types of pathogens and spoilage bacteria are prevented from multiplying to microbiologically significant levels in properly refrigerated foods that are not out of date. Therefore, it is imperative that refrigerated and frozen potentially hazardous foods be received at the temperature specified in Code.
High temperatures for a long enough time, such as those associated with thorough cooking kill or inactivate many types of microorganisms. However, cooking does not

=====
SEMANTIC | question=q4 | chunk_length=3326 chars | cosine=0.6899
=====
For this, no specific temperature is required for the consumer's protection.
Reheating for hot holding When food is held, cooled, and reheated in a food facility, there is an increased risk from contamination caused by personnel, equipment, procedures, or other factors. If food is held at improper temperatures for enough time, pathogens have the opportunity to multiply to dangerous numbers. Proper reheating provides a major degree of assurance that pathogens will be eliminated.
The potential for growth of pathogenic bacteria is greater in reheated cooked foods than in raw foods. This is because spoilage bacteria, which inhibit the growth of pathogens by competition on raw product, are killed during cooking. Subsequent recontamination will allow pathogens to grow without competition if temperature abuse occurs.
PROTECT FROM CONTAMINATION
13. Returned and re-service of food
Authority: California Retail Food Code, §§ 114079
This violation category is used to indicate the compliance status with ensuring foods which have been served to a consumer are not returned and reserved to another consumer.
IN – "IN" shall be marked when consumers are not served food items previously provided to another consumer.
N/A – "N/A" shall be marked if the facility only handles prepackaged foods.
N/O – "N/O" shall be marked if the food facility is not serving food at the time of inspection.
OUT – This violation category shall be marked MINOR; examples include:
• Re-served tortilla chips, salsa, or unpackaged bread. (VC&D) [§114079]
• Popcorn, nuts, or snack mix left on the bar for communal service. (VC&D) [§114079]
• Previously served steamed rice used in the preparation of fried rice. (VC&D) [§114079]
Notes:
☐ Containers of non-potentially hazardous foods such as ketchup, steak sauce, hot sauce, etc., can be transferred from one consumer to another if the food is protected from contamination and the container is closed between uses.
☐ Schools may provide a sharing table where food service staff, pupils, and faculty may return food items and make those food items available to others during the course of a regular school mealtime.
☐ Food in unopened original packages that are in sound condition and checked periodically on a regular basis, such as crackers or sugar, can be re-served.
Public Health Reason:
Returned food and reserving of food Food can serve as a means of person-to-person transmission of disease agents such as hepatitis A virus. Any unpackaged foods, even bakery goods in a breadbasket that are not potentially hazardous and that have been served to a consumer, but not eaten, can become vehicles for transmitting pathogenic microorganisms from the initial consumer to the next consumer if the food is served again.
14. Food in good condition, safe and unadulterated
Authority: California Retail Food Code, §§ 113967, 113976, 113988, 113982, 113988, 113990, 114035, 114041, 114254.3, 114355, 114419; Sherman Food, Drug, and Cosmetic Law, §§ 110605, 110610; California Code of Regulations, Title 17, Section 13675; California Health & Safety Code § 110611.
This violation category is used to indicate the compliance status with food kept in good condition, safe, and unadulterated.
IN – "IN" shall be marked when foods at the food facility are observed to be in good condition, safe, and unadulterated.

=====
SENTENCE-WINDOW | question=q4 | chunk_length=403 chars | cosine=0.7803
=====
☐ Foods approved for holding up to 45°F include the following: raw shell eggs; un-shucked live molluscan shellfish; pasteurized milk and pasteurized milk products in original sealed containers; PHFs held for dispensing in serving lines and salads bars during periods not to exceed 12 hours in any 24-hour period; or PHF held in vending machines; or PHF held for sampling at a Certified Farmers' Market.
```

```
reports > hw03 > RUN_LOG.txt
```

```
1 Loaded 4 document(s)
2
3 [token] 693 nodes
4 [semantic] 243 nodes
5 [sentence_window] 4749 nodes
6
7 [token/q1] query embedding dim=384, first 8=[ 0.0438 -0.0394 -0.0146 0.0076 -0.079 0.1038 -0.0642 -0.0168]
8 [token/q1] latency=37.11 ms, 3 node(s) returned
9 [token/q1] query shape=(384,), docs shape=(3, 384)
10 rank=1 store_score=0.7003 cosine=0.6488 len=1104 preview='- _____ Remaining Points: = Score The table below describes the letter grade achieved base on the score received.
11 rank=2 store_score=0.6489 cosine=0.6833 len=1059 preview='POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Closure sign (if applicable) is posted as required.'
12 rank=3 store_score=0.6346 cosine=0.6696 len=1018 preview='cities within Los Angeles County that have adopted Ordinance #97-0071, the grade/score card a facility receives at the conclusion of a routine inspection must be'
13 [token/q2] query embedding dim=384, first 8=[ 0.0378 -0.0116 -0.0398 0.0386 -0.0899 0.0595 -0.0415 -0.0405]
14 [token/q2] latency=35.03 ms, 3 node(s) returned
15 [token/q2] query shape=(384,), docs shape=(3, 384)
16 rank=1 store_score=0.7394 cosine=0.6776 len=1029 preview='ABC RETAIL FOOD INSPECTION GUIDE San Bernardino County Public Health Environmental Health Services 385 N Arrowhead Ave., 2nd floor San Bernardino, CA 92415-0160'
17 rank=2 store_score=0.7189 cosine=0.6905 len=1084 preview='Notice of Closure Card will be posted in the front window or in a location approved by the health inspector. Removal of the Notice of Closure by anyone other than the person who posted it is prohibited.'
18 rank=3 store_score=0.6860 cosine=0.6063 len=1110 preview='who is involved in the preparation, storage, or service of food in a food facility. Food Handler Card - Issued after passing an EHS approved training class. Each'
19 [token/q3] query embedding dim=384, first 8=[-0.0058 0.078 0.0472 0.042 0.0252 -0.0286 -0.0292 -0.0373]
20 [token/q3] latency=33.43 ms, 3 node(s) returned
21 [token/q3] query shape=(384,), docs shape=(3, 384)
22 rank=1 store_score=0.6799 cosine=0.5580 len=1088 preview='foods prepared it would be impossible to generalize how fast a typical product loses temperature after cooking but holding a food product for 4 hours at room temperature'
23 rank=2 store_score=0.6350 cosine=0.6431 len=889 preview='41°F within 4 hours. (§114037(a)) 20 | Page N.A. This item may be marked N.A. when the food facility does not hold hot or cold food. N.O. This item may be marked N.A. when the food facility does not hold hot or cold food.'
24 rank=3 store_score=0.6311 cosine=0.5905 len=1083 preview='Additional Violations section. \uf008 A food facility that prepares food shall not operate if the water temperature at the warewashing sink faucet is below 180°F and'
25 [token/q4] query embedding dim=384, first 8=[ 0.0204 -0.0437 0.012 0.0414 0.0115 0.0264 -0.0077 -0.0099]
26 [token/q4] latency=35.42 ms, 3 node(s) returned
27 [token/q4] query shape=(384,), docs shape=(3, 384)
28 rank=1 store_score=0.7855 cosine=0.7388 len=1237 preview='the more permissive conditions are for growth. Hot holding The hot holding of food at 135°F or greater is sufficient to prevent the growth of pathogens and is t'
29 rank=2 store_score=0.7101 cosine=0.6760 len=1234 preview='Do not rely solely upon discussions with managers or cooks to determine compliance or non-compliance. The temperature of cooked potentially hazardous foods durin'
30 rank=3 store_score=0.7096 cosine=0.7384 len=1184 preview='hot holding carts being used to transport food to buffet tables, satellite kitchens, or off-site catering events'. \uf076 As a rule, every effort should be made to a'
31 [token/q5] query embedding dim=384, first 8=[-0.0098 0.0149 0.048 0.0507 0.0126 -0.0337 -0.0122 -0.0315]
32 [token/q5] latency=40.44 ms, 3 node(s) returned
33 [token/q5] query shape=(384,), docs shape=(3, 384)
34 rank=1 store_score=0.7388 cosine=0.7720 len=930 preview='food employee is washing their hands while wearing single-use gloves or donning previously worn single-use gloves. (§113973(b)(c)) Examples of MINOR violations'
35 rank=2 store_score=0.7123 cosine=0.5635 len=1092 preview='be followed by thorough handwashing. Handwashing is a critical factor in reducing fecal-oral pathogens that can be transmitted from hands to ready-to-eat food'
36 rank=3 store_score=0.6684 cosine=0.5889 len=904 preview='the washing of hands. Because handwashing is such an important factor in the prevention of foodborne illness, sufficient facilities must be available to make ha'
37
38 [semantic/q1] query embedding dim=384, first 8=[ 0.0438 -0.0394 -0.0146 0.0076 -0.079 0.1038 -0.0642 -0.0168]
39 [semantic/q1] latency=24.03 ms, 3 node(s) returned
40 [semantic/q1] query shape=(384,), docs shape=(3, 384)
41 rank=1 store_score=0.6489 cosine=0.6833 len=4077 preview='POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Closure sign (if applicable) is posted as required.'
42 rank=2 store_score=0.6274 cosine=0.6079 len=8484 preview='Refer to the following page for an example of the OIR. 8 | Page 9 | Page Understanding Your Grade The purpose of performing a routine inspection must be'
43 rank=3 store_score=0.5861 cosine=0.5474 len=331 preview='This guide provides information on common terms used in the California Retail Food Code (Cal Code) and the San Bernardino County Public Health Environmental Health Services 385 N Arrowhead Ave., 2nd floor San Bernardino, CA 92415-0160'
44 [semantic/q2] query embedding dim=384, first 8=[ 0.0378 -0.0116 -0.0398 0.0386 -0.0899 0.0595 -0.0415 -0.0405]
45 [semantic/q2] latency=31.68 ms, 3 node(s) returned
```

```
code\hw03\run_retrieval.py : LERNO 21 no such file or directory
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code\hw03\run_retrieval.py 2>61 | tee reports\hw03\RUN_LOG.txt
Loaded 4 document(s)

[token] 693 nodes
[semantic] 243 nodes
[sentence_window] 4749 nodes

[token/q1] query embedding dim=384, first 8=[ 0.0438 -0.0394 -0.0146 0.0076 -0.079 0.1038 -0.0642 -0.0168]
[token/q1] latency=35.59 ms, 3 node(s) returned
[token/q1] query shape=(384,), docs shape=(3, 384)
rank=1 store_score=0.7003 cosine=0.6488 len=1104 preview='- _____ Remaining Points: = Score The table below describes the letter grade achieved base on the score received.
Score Grade Description 90 '
rank=2 store_score=0.6489 cosine=0.6833 len=1059 preview='POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Closure sign (if applicable) is posted as required.'
rank=3 store_score=0.6346 cosine=0.6696 len=1018 preview='cities within Los Angeles County that have adopted Ordinance #97-0071, the grade/score card a facility receives at the conclusion of a routine inspection must be'
[token/q2] query embedding dim=384, first 8=[ 0.0378 -0.0116 -0.0398 0.0386 -0.0899 0.0595 -0.0415 -0.0405]
[token/q2] latency=34.66 ms, 3 node(s) returned
[token/q2] query shape=(384,), docs shape=(3, 384)
rank=1 store_score=0.7394 cosine=0.6776 len=1029 preview='ABC RETAIL FOOD INSPECTION GUIDE San Bernardino County Public Health Environmental Health Services 385 N Arrowhead Ave., 2nd floor San Bernardino, CA 92415-0160'
rank=2 store_score=0.7189 cosine=0.6905 len=1084 preview='Notice of Closure Card will be posted in the front window or in a location approved by the health inspector. Removal of the Notice of Closure by anyone other than the person who posted it is prohibited.'
rank=3 store_score=0.6860 cosine=0.6063 len=1110 preview='who is involved in the preparation, storage, or service of food in a food facility. Food Handler Card - Issued after passing an EHS approved training class. Each'
[token/q3] query embedding dim=384, first 8=[-0.0058 0.078 0.0472 0.042 0.0252 -0.0286 -0.0292 -0.0373]
[token/q3] latency=33.23 ms, 3 node(s) returned
[token/q3] query shape=(384,), docs shape=(3, 384)
rank=1 store_score=0.6799 cosine=0.5580 len=1088 preview='foods prepared it would be impossible to generalize how fast a typical product loses temperature after cooking but holding a food product for 4 hours at room temperature'
rank=2 store_score=0.6350 cosine=0.6431 len=889 preview='41°F within 4 hours. (§114037(a)) 20 | Page N.A. This item may be marked N.A. when the food facility does not hold hot or cold food. N.O. This item may be marked N.A. when the food facility does not hold hot or cold food.'
rank=3 store_score=0.6311 cosine=0.5905 len=1083 preview='Additional Violations section. \uf008 A food facility that prepares food shall not operate if the water temperature at the warewashing sink faucet is below 180°F and'
[token/q4] query embedding dim=384, first 8=[ 0.0204 -0.0437 0.012 0.0414 0.0115 0.0264 -0.0077 -0.0099]
[token/q4] latency=33.31 ms, 3 node(s) returned
[token/q4] query shape=(384,), docs shape=(3, 384)
rank=1 store_score=0.7855 cosine=0.7388 len=1237 preview='the more permissive conditions are for growth. Hot holding The hot holding of food at 135°F or greater is sufficient to prevent the growth of pathogens and is t'
rank=2 store_score=0.7101 cosine=0.6760 len=1234 preview='Do not rely solely upon discussions with managers or cooks to determine compliance or non-compliance. The temperature of cooked potentially hazardous foods during'
rank=3 store_score=0.7096 cosine=0.7384 len=1184 preview='hot holding carts being used to transport food to buffet tables, satellite kitchens, or off-site catering events'. \uf076 As a rule, every effort should be made to a'
[token/q5] query embedding dim=384, first 8=[-0.0098 0.0149 0.048 0.0507 0.0126 -0.0337 -0.0122 -0.0315]
[token/q5] latency=35.40 ms, 3 node(s) returned
[token/q5] query shape=(384,), docs shape=(3, 384)
rank=1 store_score=0.7388 cosine=0.7720 len=930 preview='food employee is washing their hands while wearing single-use gloves or donning previously worn single-use gloves. (§113973(b)(c)) Examples of MINOR violations'
rank=2 store_score=0.7123 cosine=0.5635 len=1092 preview='be followed by thorough handwashing. Handwashing is a critical factor in reducing fecal-oral pathogens that can be transmitted from hands to ready-to-eat food'
rank=3 store_score=0.6684 cosine=0.5889 len=904 preview='the washing of hands. Because handwashing is such an important factor in the prevention of foodborne illness, sufficient facilities must be available to make ha'
[token/q6] query embedding dim=384, first 8=[ 0.0438 -0.0394 -0.0146 0.0076 -0.079 0.1038 -0.0642 -0.0168]
[semantic/q1] latency=21.13 ms, 3 node(s) returned
[semantic/q1] query shape=(384,), docs shape=(3, 384)
```

Retrieval quality comparison for Token vs. Semantic vs. Sentence-window

Token (chunk_size=256, chunk_overlap=32): chunk_size of 256 accommodates the input size for MiniLM (~256 tokens), thus nearly full embedding of each chunk. The token overlap of 32 ensures that texts close to the boundary will not be broken without any common context.

Semantic (buffer_size=1, breakpoint_percentile_threshold=95): the embed model detects topic transition. Every sentence is embedded along with the previous and next sentences, and boundaries are only created when the distance between consecutive sentences is in the top 5% of the document's distances.

Sentence-window (window_size=3): every node corresponds to one sentence, while there are 3 sentences on both sides as node metadata ("window").

```
def build_indexes(docs: list) -> tuple[dict[str, VectorStoreIndex], list[dict]]:
    token_splitter = TokenTextSplitter(chunk_size=MAX_TOKEN_CHUNK_SIZE, chunk_overlap=32)
    semantic_splitter = SemanticSplitterNodeParser(
        buffer_size=1, breakpoint_percentile_threshold=95, embed_model=embed_model
    )
    window_splitter = SentenceWindowNodeParser.from_defaults(window_size=3)

    indexes: dict[str, VectorStoreIndex] = {}
    chunk_stat_rows: list[dict] = []
    for name, splitter in [
        ("token", token_splitter),
        ("semantic", semantic_splitter),
        ("sentence_window", window_splitter),
    ]:
        nodes = splitter.get_nodes_from_documents(docs)
        print(f"[{name}] {len(nodes)} nodes")
        for i, node in enumerate[Any](nodes):
            chunk_stat_rows.append({
                "technique": name,
                "chunk_index": i,
                "chunk_length": len(node.get_content()),
            })
        indexes[name] = VectorStoreIndex(nodes, embed_model=embed_model)
    return indexes, chunk_stat_rows

def cosine_similarity(query_vec: np.ndarray, doc_vecs: np.ndarray) -> np.ndarray:
    if doc_vecs.shape[0] == 0:
        return np.array([])
    query_unit = query_vec / np.linalg.norm(query_vec)
    doc_units = doc_vecs / np.linalg.norm(doc_vecs, axis=1, keepdims=True)
    return doc_units @ query_unit

def retrieve(technique: str, index: VectorStoreIndex, question: dict, k: int) -> list[dict]:
    query = question["text"].strip()
    qid = question["id"]

    query_vec = np.array(embed_model.get_query_embedding(query))
    print(f"[{technique}/{qid}] query embedding dim={query_vec.shape[0]}, first 8={np.round(query_vec[:8], 4)}")

    start = time.perf_counter()
    retriever = index.as_retriever(similarity_top_k=k)
    result_nodes = retriever.retrieve(query)
```

Technique	Chunks	Avg chunk length (chars)	Top-1 cosine	Mean@3 cosine	Recall@3	Mean latency (ms)
Token	693	1002	0.6790	0.6564	60% (3/5)	36.28
Semantic	243	2445	0.6353	0.6233	60% (3/5)	21.57
Sentence-window	4749	125	0.6740	0.6530	40% (2/5)	154.25

B2. Indexing (in memory)

All of these three chunking strategies will have their own `VectorStoreIndex` constructed over their own set of nodes, using the default `SimpleVectorStore`, where all vectors are stored in memory and never persisted on disk.

```
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/hw03/run_retrieval.py

Loaded 4 document(s)

[token] 693 nodes
[token] index built, 693 nodes, vector_store=SimpleVectorStore
[semantic] 243 nodes
[semantic] index built, 243 nodes, vector_store=SimpleVectorStore
[sentence_window] 4749 nodes
[sentence_window] index built, 4749 nodes, vector_store=SimpleVectorStore
```

B3. Retrieval-only function

```

def retrieve(technique: str, index: VectorStoreIndex, question: dict, k: int) -> list[dict]:
    query = question["text"].strip()
    qid = question["id"]

    print(f"\n=== TECHNIQUE: {DISPLAY_NAMES[technique]} | question={qid} ===")
    print(f"Query: {query}")

    query_vec = np.array(embed_model.get_query_embedding(query))
    print(f"query embedding dim={query_vec.shape[0]}, first 8={np.round(query_vec[:8], 4)}")

    start = time.perf_counter()
    retriever = index.as_retriever(similarity_top_k=k)
    result_nodes = retriever.retrieve(query)
    latency_ms = (time.perf_counter() - start) * 1000

    chunk_texts = [n.get_content() for n in result_nodes]
    chunk_vecs = (
        np.array(embed_model.get_text_embedding_batch(chunk_texts))
        if chunk_texts
        else np.zeros((0, query_vec.shape[0]))
    )
    print(f"query shape={query_vec.shape}, docs shape={chunk_vecs.shape}")
    print(f"latency={latency_ms:.2f} ms, {len(result_nodes)} node(s) returned\n")

    cos_sims = cosine_similarity(query_vec, chunk_vecs)

    rows = []
    table_rows = []
    for rank, (node, cos_sim) in enumerate[tuple[NodeWithScore, Any]](zip[tuple[NodeWithScore, Any]](result_nodes, cos_sims), start=1):
        text = node.get_content()
        preview = text[:160].replace("\n", " ")
        window_text = node.metadata.get("window") if technique == "sentence_window" else None
        table_rows.append({
            "rank": rank,
            "store_score": round(float(node.score), 4) if node.score is not None else None,
            "cosine_sim": round(float(cos_sim), 4),
            "chunk_len": len(text),
            "preview": preview,
        })
    rows.append({
        "technique": technique,
        "question_id": qid,
        "rank": rank,
        "store_score": float(node.score) if node.score is not None else None,
        "cosine_similarity": float(cos_sim),
        "chunk_length": len(text),
        "window_length": len(window_text) if window_text else None,
        "latency_ms": round(latency_ms, 3),
        "preview": preview,
        "text": text,
    })

```

```

indexes, chunk_stat_rows = build_indexes(docs)
print()

all_rows: list[dict] = []
for technique, index in indexes.items():
    for question in questions:
        all_rows.extend(retrieve(technique, index, question, TOP_K))
    print()

jsonl_path = RAW_DIR / "retrieval_runs.jsonl"
with jsonl_path.open("w", encoding="utf-8") as f:
    for row in all_rows:
        f.write(json.dumps(row) + "\n")

csv_path = RAW_DIR / "retrieval_runs.csv"
fieldnames = list[Any](all_rows[0].keys())
with csv_path.open("w", newline="", encoding="utf-8") as f:
    writer = csv.DictWriter(f, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(all_rows)

print(f"Wrote {len(all_rows)} rows to {jsonl_path} and {csv_path}")

chunk_stats_path = RAW_DIR / "chunk_stats.csv"
with chunk_stats_path.open("w", newline="", encoding="utf-8") as f:
    writer = csv.DictWriter(f, fieldnames=["technique", "chunk_index", "chunk_length"])
    writer.writeheader()
    writer.writerows(chunk_stat_rows)

```

Compute the query embedding (show its dimension and the first 8 values).

```

def cosine_similarity(query_vec: np.ndarray, doc_vecs: np.ndarray) -> np.ndarray:
    if doc_vecs.shape[0] == 0:
        return np.array([])
    query_unit = query_vec / np.linalg.norm(query_vec)
    doc_units = doc_vecs / np.linalg.norm(doc_vecs, axis=1, keepdims=True)
    return doc_units @ query_unit

```

```

(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/hw03/run_retrieval.py 2>&1 | tee reports/hw03/RUN_LOG.txt
Loaded 4 document(s)
[token] 693 nodes
[token] index built, 693 nodes, vector_store=SimpleVectorStore
[semantic] 243 nodes
[semantic] index built, 243 nodes, vector_store=SimpleVectorStore
[sentence_window] 4749 nodes
[sentence_window] index built, 4749 nodes, vector_store=SimpleVectorStore

=== TECHNIQUE: Token | question=q1 ===
Query: According to the LA County Retail Food Inspection Guide, what score range earns a food facility a "B" letter grade, and what happens instead of a letter grade if the score is 69 or below?
Query embedding dimension: 384
First 8 values: [0.043800000101327896, -0.039400000125169754, -0.014600000344216824, 0.007600000128149986, -0.07900000363588333, 0.10379999876022339, -0.06419999897480011, -0.01679999940097332]

```

Retrieve top-k nodes; for each, compute and print

```
reports > hw03 > raw > {} retrieval_runs.jsonl

1 {"technique": "token", "question_id": "q1", "rank": 1, "store_score": 0.7002996304722631, "cosine_similarity": 0.6488123466084393, "chunk_length": 1104, "wi
2 {"technique": "token", "question_id": "q1", "rank": 2, "store_score": 0.648912059998736, "cosine_similarity": 0.6833225416553643, "chunk_length": 1059, "wi
3 {"technique": "token", "question_id": "q1", "rank": 3, "store_score": 0.6346450358416436, "cosine_similarity": 0.6696143032039985, "chunk_length": 1018, "wi
4 {"technique": "token", "question_id": "q2", "rank": 1, "store_score": 0.7394482548886978, "cosine_similarity": 0.6775545231147125, "chunk_length": 1029, "wi
5 {"technique": "token", "question_id": "q2", "rank": 2, "store_score": 0.7188564028222282, "cosine_similarity": 0.69046998290416, "chunk_length": 1084, "wind
6 {"technique": "token", "question_id": "q2", "rank": 3, "store_score": 0.6859661778038839, "cosine_similarity": 0.6062873695687092, "chunk_length": 1110, "wi
7 {"technique": "token", "question_id": "q3", "rank": 1, "store_score": 0.6798702631633442, "cosine_similarity": 0.5580102675768057, "chunk_length": 1088, "wi
8 {"technique": "token", "question_id": "q3", "rank": 2, "store_score": 0.6350296540214728, "cosine_similarity": 0.6430664945650464, "chunk_length": 889, "win
9 {"technique": "token", "question_id": "q3", "rank": 3, "store_score": 0.6310749881822323, "cosine_similarity": 0.59054438107632, "chunk_length": 1083, "wind
10 {"technique": "token", "question_id": "q4", "rank": 1, "store_score": 0.7855028225650195, "cosine_similarity": 0.7388036286867461, "chunk_length": 1237, "wi
11 {"technique": "token", "question_id": "q4", "rank": 2, "store_score": 0.710102516306247, "cosine_similarity": 0.6760397684307948, "chunk_length": 1234, "win
12 {"technique": "token", "question_id": "q4", "rank": 3, "store_score": 0.7095906497086475, "cosine_similarity": 0.7384483230115864, "chunk_length": 1184, "wi
13 {"technique": "token", "question_id": "q5", "rank": 1, "store_score": 0.7387709931198365, "cosine_similarity": 0.771961954202486, "chunk_length": 930, "wind
14 {"technique": "token", "question_id": "q5", "rank": 2, "store_score": 0.712324837380059, "cosine_similarity": 0.5634699891948609, "chunk_length": 1092, "win
15 {"technique": "token", "question_id": "q5", "rank": 3, "store_score": 0.6684367479383917, "cosine_similarity": 0.5889469418756493, "chunk_length": 904, "win
16 {"technique": "semantic", "question_id": "q1", "rank": 1, "store_score": 0.648912059998736, "cosine_similarity": 0.6833225416553643, "chunk_length": 4077, "
17 {"technique": "semantic", "question_id": "q1", "rank": 2, "store_score": 0.6274488793454139, "cosine_similarity": 0.6078584753984086, "chunk_length": 8484, "
18 {"technique": "semantic", "question_id": "q1", "rank": 3, "store_score": 0.5861128573431955, "cosine_similarity": 0.5474497443781864, "chunk_length": 331, "
19 {"technique": "semantic", "question_id": "q2", "rank": 1, "store_score": 0.7394482548886978, "cosine_similarity": 0.6775545231147125, "chunk_length": 693, "
20 {"technique": "semantic", "question_id": "q2", "rank": 2, "store_score": 0.6891836887103913, "cosine_similarity": 0.6414778742519917, "chunk_length": 1624, "
21 {"technique": "semantic", "question_id": "q2", "rank": 3, "store_score": 0.6843847055940752, "cosine_similarity": 0.6314785541508581, "chunk_length": 331, "
22 {"technique": "semantic", "question_id": "q3", "rank": 1, "store_score": 0.6320426069017229, "cosine_similarity": 0.5704681731491924, "chunk_length": 4492, "
23 {"technique": "semantic", "question_id": "q3", "rank": 2, "store_score": 0.6211233098466798, "cosine_similarity": 0.6234769473531705, "chunk_length": 3696, "
24 {"technique": "semantic", "question_id": "q3", "rank": 3, "store_score": 0.6094258034298498, "cosine_similarity": 0.5044213774636024, "chunk_length": 398, "
25 {"technique": "semantic", "question_id": "q4", "rank": 1, "store_score": 0.7581526889642208, "cosine_similarity": 0.6899208093670051, "chunk_length": 3326, "
26 {"technique": "semantic", "question_id": "q4", "rank": 2, "store_score": 0.710102516306247, "cosine_similarity": 0.6760397684307948, "chunk_length": 4583, "
27 {"technique": "semantic", "question_id": "q4", "rank": 3, "store_score": 0.6938457375211279, "cosine_similarity": 0.7532280851145899, "chunk_length": 398, "
28 {"technique": "semantic", "question_id": "q5", "rank": 1, "store_score": 0.6436488497001711, "cosine_similarity": 0.555808384402868, "chunk_length": 586, "

```

```
reports > hw03 > raw > retrieval_runs.csv > data

1 technique,question_id,rank,store_score,cosine_similarity,chunk_length>window_length,latency_ms,preview,text
2 token,q1,1,0.7002996304722631,0.6488123466084393,1104,32.71,- Remaining Points: = Score The table below describes the letter grade achieved base on the s
3 Score Grade Description 90 to 100 points A Generally superior in food handling practices and overall food facility maintenance and sanitation.
4 80 to 89 points B Generally acceptable in food handling practices and overall food facility maintenance and sanitation.
5 70 to 79 points C Generally unacceptable in food handling practices and overall general food facility maintenance and sanitation (A re-score inspection is required).
6 0 to 69 points Closure Poor food handling practices and overall food facility maintenance and sanitation.
7 At the conclusion of the inspection, the Health Inspector will place the Letter Grade Card in the facility's window or other approved location. The Letter Grade Card must not be d
8
9 10 | Page
10 The following table lists the requirements associated with each letter grade.
11 Grade Requirement A There will be no need for additional inspections.
12 B There will be no re-score inspection"
13 token,q1,2,0.648912059998736,0.6833225416553643,1059,32.71,"POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Clos
14 An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Closure sign (if applicable) is posted as required at food facilities located in uninc
15 A food facility will be issued one of the following by the EHS at the time of inspection:
16 Placard Type Description "A" Letter Grade Final routine inspection score between 90 100 "B" Letter Grade Final routine inspection score between 80 89 "C" Letter Grade Final ro
17 token,q1,3,0.6346450358416436,0.6696143032039985,1018,32.71,"cities within Los Angeles County that have adopted Ordinance #97-0071, the grade/score card a facility receives at th
18 48. Permits Available
19 Authority: California Retail Food Code, §§ 114381, 114387 Col 10: text
20 This violation category is used to indicate the compliance status with permit requirements.
21 OUT "OUT" shall be marked if a food facility does not have a valid public health permit (PHP) and/or is operating outside the scope of their permit.
22 Examples of violations include:
23 • A food facility is open for business and does not have a valid PHP. [§114381(a), §114387]
24 • Food facility is handling/storing food beyond the scope of the issued permit (e.g., liquor store preparing food). (Refer to the Plan Check program)"
25 token,q2,1,0.7394482548886978,0.6775545231147125,1029,33.586,"ABC RETAIL FOOD INSPECTION GUIDE San Bernardino County Public Health Environmental Health Services 385 N Arrowhead A
26 San Bernardino County Public Health Environmental Health Services 385 N Arrowhead Ave., 2nd floor San Bernardino, CA 92415-0160 1-800-442-2283 Fax: 909-387-4323
27 Public Health Environmental Health Services http://wp.sbcounty.gov/dph/programs/ehs/

```

Print the shapes of the query vector and the stacked doc vectors

```
=== TECHNIQUE: Token | question=q1 ===
Query: According to the LA County Retail Food Inspection Guide, what score range earns a food facility a "B" letter grade, and what happens instead of a letter grade if the score is 69 or below
?
query embedding dim=384, first 8=[ 0.0438 -0.0394 -0.0146 0.0076 -0.079 0.1038 -0.0642 -0.0160]
query shape=(384,), docs shape=(3, 384)
latency=33.92 ms, 3 node(s) returned

rank store_score cosine_sim chunk_len
preview
1 0.7003 0.6488 1104 - Remaining Points: = Score The table below describes the letter grade achieved base on the score received. Score Grade Descri
ption 90
2 0.6489 0.6833 1059 POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Closure sign (if applicable) is poste
d as requ
3 0.6346 0.6696 1018 cities within Los Angeles County that have adopted Ordinance #97-0071, the grade/score card a facility receives at the conclusion of a routine inspecti
on must b

```

```
=== TECHNIQUE: Semantic | question=q1 ===
Query: According to the LA County Retail Food Inspection Guide, what score range earns a food facility a "B" letter grade, and what happens instead of a letter grade if the score is 69 or below
?
query embedding dim=384, first 8=[ 0.0438 -0.0394 -0.0146 0.0076 -0.079 0.1038 -0.0642 -0.0160]
query shape=(384,), docs shape=(3, 384)
latency=19.39 ms, 3 node(s) returned

rank store_score cosine_sim chunk_len
preview
1 0.6489 0.6833 4077 POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Closure sign (if applicable) is poste
d as requ
2 0.6274 0.6079 8484 Refer to the following page for an example of the OIR. 8 | Page 9 | Page Understanding Your Grade The purpose of performing food facility inspections
is to pr
3 0.5861 0.5474 331 This guide provides information on common terms used in the California Retail Food Code (Cal Code) and the San Bernardino County Code (SBCC) to assist
in unders

```

```

=== TECHNIQUE: Sentence-window | question=q1 ===
Query: According to the LA County Retail Food Inspection Guide, what score range earns a food facility a "B" letter grade, and what happens instead of a letter grade if the score is 69 or below ?
query embedding dim=384, first 8=[ 0.0438 -0.0394 -0.0146 0.0076 -0.079 0.1038 -0.0642 -0.0168]
query shape=(384,), docs shape=(3, 384)
latency=160.69 ms, 3 node(s) returned

rank  score_score cosine_sim chunk_len
preview
1 0.6308 0.5759 288 POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Closure sign (if applicable) is posted as required.
2 0.6297 0.6325 143 Score Grade Description 90 to 100 points A Generally superior in food handling practices and overall food facility maintenance and sanitation.
3 0.6227 0.5880 100 • Received a score letter of "B" or lower (89 or less) on two of the last three graded inspections.

```

```
(vnm) (base) pc@pcs-MacBook-Pro data268-7117 % ls -la reports/hw03/raw/
head -5 reports/hw03/raw/retrieval_runs.csv
total 840
drwxr-xr-x@ 6 pc staff 192 Sep 21 02:25 .
drwxr-xr-x@ 12 pc staff 384 Sep 21 03:22 ..
-rw-r--r--@ 1 pc staff 0 Sep 20 23:01 .gitkeep
-rw-r--r--@ 1 pc staff 135002 Sep 21 12:57 chunk_stats.csv
-rw-r--r--@ 1 pc staff 85589 Sep 21 12:57 retrieval_runs.csv
-rw-r--r--@ 1 pc staff 94685 Sep 21 12:57 retrieval_runs.json
-rw-r--r--@ 1 pc staff 11614 Sep 21 12:57 retrieval_runs.txt
technique,question_id,rank,score,similarity,chunk_length,window_length,latency_ms,preview,token,1,1,0.7082996304722631,0.6488123466041047,1104,3.923,- Remaining Points: = Score The table below describes the letter grade achieved base on the score received
Score Grade Description 90 "-" Remaining Points: = Score The table below describes the letter grade achieved base on the score received
Score grade description to points A Generally superior in food handling practices and overall food facility maintenance and sanitation.
90 to 99 points B Generally acceptable in food handling practices and overall food facility maintenance and sanitation.
70 to 79 points C Generally unacceptable in food handling practices and overall general food facility maintenance and sanitation (A re-score inspection is required).
```

```

def main() -> None:
    runs = pd.read_csv(RAW_DIR / "retrieval_runs.csv")
    runs = runs.dropna(subset=["rank"]) # drop the "no nodes returned" placeholder rows, if any
    chunk_stats = pd.read_csv(RAW_DIR / "chunk_stats.csv")
    answer_keywords = load_answer_keywords()

    chunks_summary = chunk_stats.groupby("technique")["chunk_length"].agg(
        chunks="count", avg_chunk_length="mean"
    )

    top1_cosine = (
        runs[runs["rank"] == 1].groupby("technique")["cosine_similarity"].mean().rename("top1_cosine")
    )
    mean_at_k_cosine = runs.groupby("technique")["cosine_similarity"].mean().rename("mean_at_k_cosine")

    # latency_ms is repeated across a question's k rows (one retrieve()
    # call produces it) - drop duplicates per (technique, question) before
    # averaging, so a question with fewer returned nodes isn't overweighted.
    per_question_latency = runs.drop_duplicates(subset=["technique", "question_id"])
    mean_latency = per_question_latency.groupby("technique")["latency_ms"].mean().rename("mean_latency_ms")

    recall_at_k = compute_recall_at_k(runs, answer_keywords).rename("recall_at_k")

    summary = pd.concat(
        [chunks_summary, top1_cosine, mean_at_k_cosine, recall_at_k, mean_latency], axis=1
    )
    summary = summary.reindex(["token", "semantic", "sentence_window"])

    print(summary.to_string(float_format=lambda x: f"{x:.4f}"))

    display_names = {
        "token": "Token",
        "semantic": "Semantic",
        "sentence_window": "Sentence-window",
    }

```

```
reports > hw03 > RUN_LOG.txt
1  === RUN START: 2026-09-21T20:57:41.336959+00:00 ===
2
3  Loaded 4 document(s)
4
5  [token] 693 nodes
6  [token] index built, 693 nodes, vector_store=SimpleVectorStore
7  [semantic] 243 nodes
8  [semantic] index built, 243 nodes, vector_store=SimpleVectorStore
9  [sentence_window] 4749 nodes
10 [sentence_window] index built, 4749 nodes, vector_store=SimpleVectorStore
11
12
13 === TECHNIQUE: Token | question=q1 ===
14 Query: According to the LA County Retail Food Inspection Guide, what score range earns a food facility a "B" letter grade, and what happens instead of a letter grade if the score
15 Query embedding dimension: 384
16 First 8 values: [0.043800000101327896, -0.039400000125169754, -0.014600000344216824, 0.007600000128149986, -0.07900000363588333, 0.10379999876022339, -0.06419999897480011, -0.0167
17 query shape=(384,), docs shape=(3, 384)
18 latency=36.58 ms, 3 node(s) returned
19
20 rank store_score cosine_sim chunk_len
21 1 0.7003 0.6488 1104 - Remaining Points: = Score The table below describes the letter grade achieved base on the score received. Score
22 2 0.6489 0.6833 1059 POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card, or Notice of Closure sign (if applic
23 3 0.6346 0.6696 1018 cities within Los Angeles County that have adopted Ordinance #97-0071, the grade/score card a facility receives at the conclusion of a re
24
25 === TECHNIQUE: Token | question=q2 ===
26 Query: Per the San Bernardino County ABC Retail Food Inspection Guide, what must a food facility that receives a "C" grade do to avoid closure, and what phone number should an ope
27 Query embedding dimension: 384
28 First 8 values: [0.0379999911785126, -0.011599999852478504, -0.039799999445678004, 0.038600001484155655, -0.08990000188350677, 0.05950000137090683, -0.04149999842047691, -0.04050
29 query shape=(384,), docs shape=(3, 384)
30 latency=40.58 ms, 3 node(s) returned
31
32 rank store_score cosine_sim chunk_len
33 1 0.7394 0.6776 1029 ABC RETAIL FOOD INSPECTION GUIDE San Bernardino County Public Health Environmental Health Services 385 N Arrowhead Ave., 2nd floor San Be
34 2 0.7189 0.6905 1084 Notice of Closure Card will be posted in the front window or in a location approved by the health inspector. Removal of the Notice of Clo
35 3 0.6860 0.6063 1110 who is involved in the preparation, storage, or service of food in a food facility. Food Handler Card – Issued after passing an EHS appro
36
37 === TECHNIQUE: Token | question=q3 ===
38 Query: In the inspection report for Dat Thanh Restaurant (Est ID FA0003611, 5035 Fruitridge Rd, Sacramento), what specific cooling violation did the inspector observe, and what te
39 Query embedding dimension: 384
40 First 8 values: [-0.00579999926239252, 0.07800000160932541, 0.047200001776218414, 0.041999999433755075, 0.025200000032782555, -0.028599999845027924, -0.029200000680433647, -0.037
41 query shape=(384,), docs shape=(3, 384)
42 latency=40.28 ms, 3 node(s) returned
43
44 rank store_score cosine_sim chunk_len
45 1 0.6799 0.5580 1088 foods prepared it would be impossible to generalize how fast a typical product loses temperature after cooking but holding a food product
46 2 0.6350 0.6431 889 41°F within 4 hours. (§114037(a)) 20 | Page N.A. This item may be marked N.A. when the food facility does not hold hot or cold food. N.
47 3 0.6311 0.5905 1083 Additional Violations section. □ A food facility that prepares food shall not operate if the water temperature at the warewashing sink fa
48
49 === TECHNIQUE: Token | question=q4 ===
50 Query: What are the required hot-holding and cold-holding temperatures for potentially hazardous food (PHF), and what exception lets some foods be held up to 45F?
51 Query embedding dimension: 384
52 First 8 values: [0.024000000736117363, -0.0436999998408555984, 0.012000000104308128, 0.0414000004529953, 0.011500000022351742, 0.026399999856948853, -0.007699999958276749, -0.00989
53 query shape=(384,), docs shape=(3, 384)
54 latency=37.34 ms, 3 node(s) returned
55
```

```
1
2  === TECHNIQUE: Semantic | question=q1 ===
3  Query: According to the LA County Retail Food Inspection Guide, what score range earns a food facility a "B" letter grade, and what happens ins
4  Query embedding dimension: 384
5  First 8 values: [0.043800000101327896, -0.039400000125169754, -0.014600000344216824, 0.007600000128149986, -0.07900000363588333, 0.103799998760
6  query shape=(384,), docs shape=(3, 384)
7  latency=22.94 ms, 3 node(s) returned
8
9  rank store_score cosine_sim chunk_len
10 1 0.6489 0.6833 4077 POSTING REQUIREMENTS An EHS shall verify at the time of an inspection that a Grade Card, Score Card,
11 2 0.6274 0.6079 8484 Refer to the following page for an example of the OIR. 8 | Page 9 | Page Understanding Your Grade T
12 3 0.5861 0.5474 331 This guide provides information on common terms used in the California Retail Food Code (Cal Code) an
13
```

Retrieval quality comparison — Token vs. Semantic vs. Sentence-window

Technique	Chunks	Avg chunk length (chars)	Top-1 cosine	Mean@3 cosine	Recall@3	Mean latency (ms)
Token	693	1002	0.6790	0.6564	60% (3/5)	34.71
Semantic	243	2445	0.6353	0.6233	60% (3/5)	22.06
Sentence-window	4749	125	0.6740	0.6530	40% (2/5)	152.00

Observations

Best performance was seen from Token chunking, with the highest top-1 cosine (0.6790) and mean@3 cosine (0.6564), as well as tied for best Recall@3 with Semantic at 60%. Chunks of about 1,002 characters, or roughly 250 tokens (check chunk_size), are small enough to be fully embedded in under all-MiniLM-L6-v2's 256 token constraint.

Very large, few chunks were created by Semantic chunking (243 chunks, averaging 2,445 characters). The tail of these chunks likely exceeds the capacity of the model and thus does not appear in the vector output, which explains the low cosines seen here (0.6353 top-1 and 0.6233 mean@3). This was the quickest method (21.57 ms).

Sentence-window was close to Token when it came to cosine (0.6740 for top-1), but it had the lowest Recall@3 (40%) and the highest latency (154.25 ms). Single sentences (around 125 characters long) fit the topic well but don't have the complete answer phrase [recall tests the selected sentence, not its window]. Also, Sentence-window uses 4,749 windows to check, and that is probably why the latency is high. With just five queries, each of them shifts recall by 20%, so the differences are relative.

Conclusion

For this dataset, token chunking is the better method. It has the best top-1 and mean@3 cosine, the best Recall@3 (60%) score tied, and it has a chunk size equal to the limit of the embedding model. The semantic method comes in second place in recall and fastest in latency, while sentence-window has the worst recall and latency.

1. What I used an AI assistant for, and what I did myself

I used Claude as a planning and debugging assistant. It helped me:

- break the session-management and RAG-corpus work into ordered steps;
- map HttpOnly/Secure/SameSite onto Starlette's SessionMiddleware;
- design the idle-timeout/logout logic (a server-side table of active sessions, keyed by a sid inside the signed cookie);
- identify what each downloaded PDF actually was (county, guide vs. individual inspection report) from its metadata/content;
- design a Recall@k check using a literal per-question keyword instead of the paraphrased expected answer;

- fix an install problem: PyTorch has no wheel for my Intel macOS + Python 3.13 machine, so sentence-transformers/HuggingFaceEmbedding couldn't install; it helped design a fastembed (ONNX) replacement with the same interface;
- To write the report files.

I did myself: sourced the dataset documents and their URLs/access dates; wrote questions.yaml and its expected answers from the documents, committed before any retrieval run; ran the three chunking pipelines and the metrics script; inspected flagged high-cosine misses by hand; took the screenshots. All numbers in the report come from my own runs, saved in RUN_LOG.txt and raw/.

2. AI output that was wrong or unsuitable, and what I independently verified

Unsuitable (setup): the first embedding setup followed the assignment literally (HuggingFaceEmbedding, needs sentence-transformers→torch). Failed: pip install torch==2.2.2 → ERROR: No matching distribution found for torch - no PyTorch wheel exists for Intel x86_64 macOS on Python 3.13.

Unsuitable (sessions): Starlette's default SessionMiddleware keeps the session in a signed cookie with no server-side record, so logout only clears the current response's cookie - a copied old cookie still passes validation. I caught this myself: replaying a saved cookie with curl after logout returned 200 from /dashboard instead of a redirect.

Independently verified:

- Embedding replacement: confirmed 384-dim output (matches the all-MiniLM-L6-v2 model card), and that two on-topic sentences score far higher (cosine 0.82) than an unrelated one (cosine 0.0006).
- Retrieval results: didn't treat cosine score as correctness - checked whether the retrieved chunk's actual text contained the answer.

3. How I detected the problems and verified the results

- **Install problem:** pip install torch==2.2.2 failed with "No matching distribution found for torch"; confirmed the cause was the missing macOS x86_64 build for this Python version.
- **Session problem:** logged in with curl -c, saved the cookie, logged out, replayed the old cookie against /dashboard - before the fix: 200; after adding the server-side sid table: 302 → /login. Repeated after the idle timeout expired (temporarily set to 3s): 200 immediately after login, 302 → /login after the timeout.
- **Retrieval quality:** Recall@3 is a string check - a hit means a literal answer_keyword (copied verbatim from the source document, not the paraphrased expected_answer)

appears in any top-3 chunk's text. Used a verbatim keyword from the start, since the paraphrase would never match literally.

- **Confident-wrong retrieval:** filtered raw/retrieval_runs.csv for cosine ≥ 0.7 rows missing the keyword. Top example: sentence_window, question q4, rank 1, cosine 0.78, from retail-food-inspection-guide.txt - the chunk is genuinely about hot/cold holding (the 41-45°F exception) but doesn't contain the literal "135°F" keyword, so it's scored a miss despite being topically relevant.

4. What I changed and why it works now

- **Embeddings:** replaced HuggingFaceEmbedding with FastEmbedONNXEmbedding (code/hw03/embeddings.py), wrapping fastembed. Runs the same all-MiniLM-L6-v2 weights on ONNXRuntime instead of PyTorch - same model, same embedding space, no torch. All three chunkers use it.
- **Cookie verification:** confirmed httponly/samesite=lax/secure directly in the raw Set-Cookie header via curl -i, and confirmed a real browser (Chromium via Playwright) actually stores and sends it - Chrome/Firefox treat localhost as a secure context, so a Secure cookie is testable there over plain HTTP.
- **Corpus cleaning:** code/hw03/prepare_corpus.py rejoins PDF-extracted lines hard-wrapped mid-sentence, and strips lines (headers/footers/page numbers) repeating across a threshold fraction of a document's pages.
- **Recall check:** Recall@3 defined on a verbatim answer_keyword per question (questions.yaml), computed reproducibly by code/hw03/compute_metrics.py.